

1 Chap 5 Source Coding

Discrete memoryless source: symbols emitted by the source are independent over time

$$I_s(k) = lb\left(\frac{1}{p_k}\right) = -lb(p_k)$$

$$H(S) = \sum_{k=0}^K p_k lb\left(\frac{1}{p_k}\right)$$

Codeword length:

$$\bar{L} = \sum_{k=0}^K \sum p_k L_k$$

$$\bar{L} \geq H(S)$$

and the equality can be true if p_k are all powers of 2 (and sum up to 1)

Or the difference can be made arbitrarily small by code extension

$$\eta = \frac{H(S)}{\bar{L}} \rightarrow 1$$

as $n \rightarrow \infty$

The unit of $\bar{L}$ and the $H(S)$ is **bits/symbol** So dont forget to check if code extension is used (eg x bits per y symbols is actually x/y bits/symbol)

A code is called uniquely decodable if all its extensions are non-singular (different) for all n

A prefix code is a code where no extensions of codeword is the prefix of other codeword

All prefix codes are uniquely decodable For discrete memoryless source S,

$$H(S^n) = nH(S), n \geq 2.$$

2 CHapter 6 CHannel coding

Binary addition is equivalent to 'XOR' operation (no carry bits)

2.1 Block channel coding

Block channel coding: each block of k message bits are encoded as a codeword consisting of n coded bits, k<n

Code rate:

$$r = k/n < 1$$

denoted as an (n, k) block code

WARNING: k is the number of bits of message. There are 2^k codewords

2.2 Hamming Distance

Hamming distance (of different bits)

$$d(c_i, c_j) = w(c_i + c_j)$$

Minimum hamming distance

$$d_{min} = \min_{\forall i \neq j} d(c_i, c_j)$$

2.3 Error detection

For an (n, k) block code with $d_{min} \geq t + 1$, it can detect up to t errors per codeword

Undetectable error probability

$$P_{ud} \leq \sum_{i=t+1}^n \binom{n}{i} p^i (1-p)^{n-i}, \quad t = d_{min} - 1, \quad \binom{n}{i} = \frac{n!}{(n-i)! i!}$$

note that this probability is upperbound (worst case)

2.4 Error Correction

For an (n, k) block code with $d_{min} \geq 2t + 1$, it can correct up to t errors per codeword by minimum-distance decoding

$$P_e \leq \sum_{i=t+1}^n \binom{n}{i} p^i (1-p)^{n-i}, \quad t = \left\lfloor \frac{d_{min} - 1}{2} \right\rfloor$$

Upper bound probability

2.5 Linear Block Code

Linear Block Code is a subset of block code for which the (binary) addition of any two codewords must be another codeword • For a linear block code, c = 0 (all-zero codeword) is a codeword • For linear block code, Hamming distance between two different codewords must be the weight of another codeword, i.e., • Thus, the minimum distance of a linear block code is given by the min of weight of code word

2.6 Systematic Code

An (n, k) linear block code is called systematic if each codeword c of length n is generated as parity bits _ message bits where each parity bit is generated by parity bit equation

2.7 Capacity of memless BSC

The maximum code rate (raw data rate) that can be "reliably" transmitted over the memoryless BSC with transition probability p is

$$r < C = 1 - H(p)$$

1. n total Permutation $n!$
2. k-permutation from n : $nPk = \frac{n!}{(n-k)!}$
3. combination of n from k $\binom{n}{k} = \frac{n!}{(n-k)!(k!)}$
4. partition/ multinomial coefficient / not all distinct permutations : $\binom{n}{m_1, m_2, m_3, \dots} = \frac{n!}{m_1! m_2! m_3! \dots}$
5. pascal identity $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$
6. binomial theorem

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$$

$$\text{so } \sum_{k=0}^n \binom{n}{k} = 2^n, \sum_{k=0}^n \binom{n}{k} (-1)^k = 0$$

7. the equation $x_1 + x_2 + \dots + x_r = n, x_i > 0 \forall i$ has $\binom{n-1}{r-1}$ positive integer solutions
8. the equation $x_1 + x_2 + \dots + x_r = n, x_i \geq 0 \forall i$ has $\binom{n+r-1}{r-1}$ nonnegative integer solutions
9. vandermonde's identity

$$\binom{m+n}{r} = \sum_{k=0}^r \binom{m}{k} \binom{n}{r-k}$$

10. hockey stick identity

$$\sum_{k=r}^n \binom{k}{r} = \binom{n+1}{r+1}$$

11. derangement

$$!n = n! \sum_{i=0}^n \frac{(-1)^i}{i!}$$

3 Chap 2 Conditional Probability and Bayes Rule

1. $P(A|B) = \frac{P(AB)}{P(B)}$ if $P(B) \neq 0$
2. $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$ if $P(B) \neq 0$
3. law of total probability

$$P(A) = \sum_i P(A|E_i)P(E_i)$$

where E_i forms a partition of the sample space (mutually exclusive and union is S)

4. Two events A and B are independent iff $P(AB) = P(A)P(B)$ In conditionnal probability, if $P(B|A) = P(A)$, $P(A) \neq 0$

3.1 Markov Chain

1. probability to absorption states to reach a specific absorption state s, the probabilities from each state $a_1, a_2, \dots, a_m$ are unique solution to $a_s = 1, a_i = 0 \forall \text{absorption state } i$ and $a_i = \sum_{j=1}^M a_j p_{ij}$
2. expected time to absorption

$$t_i = 1 + \sum_{j=1}^M t_j p_{ij}$$

where the 1 is added for each step increment

4 Chap 3 Discrete Random Variable

1. the PMF of a random variabe X is

$$P_X(k) = \begin{cases} P(X=k), & \text{if } k \in S_x, \\ 0, & \text{otherwise} \end{cases}$$

where S_x is the support of X

2. properties of PMF
 - (a) $0 \leq P_X(i) \leq 1$
 - (b) $\sum_{i=0}^{\infty} P_X(i) = 1$

3. CDF $F_X(x) = P(X \leq x)$ for $x \in \mathbb{R}$

4. expectation

$$E[X] = \sum_{\text{all } x} x P_X(x)$$

Expectations are linear, regardless of independencies

5. Moments $E[X^n] = \sum_{\text{all } x} x^n P_X(x)$ is called the n^{th} moment of X $E(X^n) \geq [E(X)]^n$

(Jensen's inequality)

6. Variance and standard deviation

$$Var[X] = E((X - E[X])^2) = E[X^2] - E[X]^2$$

$$\sigma_X = \sqrt{Var[X]}$$

$$Var[aX + b] = a^2 Var[X]$$

$Var[X+Y] = Var[X] + Var[Y]$ iff X and Y are independent

4.1 Discrete Distribution.

1. Bernoulli

PMF: $P(X = 1) = p, P(X = 0) = 1 - p$

$$\text{CDF: } F(x) = \begin{cases} 0 & x < 0 \\ 1 - p & 0 \leq x < 1 \\ 1 & x \geq 1 \end{cases}$$

Mean : p , Variance: $p(1 - p)$

2. Binomial PMF : $\binom{n}{k} p^k (1 - p)^{n-k}$

$$\text{CDF: } F(k) = \sum_{i=0}^k \binom{n}{i} p^i (1 - p)^{n-i}$$

Mean : np , Variance: $np(1 - p)$

3. Uniform PMF: $\frac{1}{b-a+1}$

$$\text{CDF: } F(x) = \frac{x-a+1}{b-a+1}$$

Mean: $\frac{a+b}{2}$, Variance: $\frac{(b-a+1)^2-1}{12}$

4. Geometric PMF: $(1 - p)^k p$

$$\text{CDF: } F(k) = 1 - (1 - p)^{k+1}$$

Mean : $\frac{1}{p}$, Variance: $\frac{1-p}{p^2}$ note that for geometric distribution $P(X > k) = (1 - p)^k$

5. Poisson PMF: $\frac{e^{-\lambda} \lambda^k}{k!}$

$$\text{CDF: } F(k) = e^{-\lambda} \sum_{i=0}^k \frac{\lambda^i}{i!}$$

Mean: λ , Variance: λ

6. Hypergeometric(N,K,n)

$$\text{PMF: } \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

Mean: $\frac{nK}{N}$, Variance: $\frac{nK(N-K)(N-n)}{N^2(N-1)}$

5 Chap 4. Multiple Discrete Random Variable

5.0.1 Joint PMF

(X, Y) with support $S_{X,Y} = \{(x_i, y_j), i, j = 1, 2, \dots\}$
 $\forall (x, y) \in S_{X,Y}, A := \{X = x\} \cap \{Y = y\}$
 $P_{X,Y}(x, y) := P(A) = P(X = x, Y = y)$

5.0.2 Marginal PMF

$$P(X = x_i) = \sum_{y_j \in S_Y} P_{X,Y}(x_i, y_j)$$

5.0.3 Expected value

$$E[g(X, Y)] = \sum_{x \in S_X} \sum_{y \in S_Y} g(x, y) P_{X,Y}(x, y)$$

5.0.4 Independence

Necessary and Sufficient Condition

$$P_{X,Y}(x, y) = P_X(x)P_Y(y) \quad \forall (x, y) \in S_{X,Y}$$

Deterministic functions of independent RVs are Independent

Independence $\rightarrow$ Correlation=0

5.1 Covariance and Correlation (same for Continuous)

$$\text{Cov}[X, Y] := E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y]$$

$$\text{Cov}[X, X] = \text{Var}[X]$$

$$\rho_{X,Y} = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y} \in [-1, 1] \text{ where } \sigma_X = \sqrt{\text{Var}[X]}$$

5.2 Sum of Multiple RV

5.2.1 Sum as Discrete Convolution

$$Z = X + Y \quad P_Z(j) = \sum_{k \in S_X, j-k \in S_Y} P_X(k)P_Y(j-k)$$

one of the distribution is flipped before shifting by j units

5.2.2 Expectation Linearity

$$E[\sum_{i=1}^n X_i] = \sum_{i=1}^n E[X_i] \text{ regardless of } X_i \text{ independence or not}$$

5.2.3 Variance

$$\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y] + 2\text{Cov}(X, Y)$$

$$\text{Var}[\sum_{i=1}^N X_i] = \sum_{i=1}^N \text{Var}[X_i] + \sum_{\{(i,j)|i \neq j\}} \text{Cov}(X_i, X_j)$$

note that i, j are not in increasing sequence and are interchangeable

5.2.4 Reduction of Mean Variance through Independent Sampling

$$S_n = \sum_{i=1}^n X_i$$

$$\text{Var}[S_n] \propto n$$

$$\bar{X} = \frac{S_n}{n}$$

$$\text{Var}[\bar{X}] = \text{Var}[\frac{S_n}{n}] = \frac{\sigma^2}{n} < \sigma^2$$

6 Continuous Random Variable

6.1 PDF

$$P(X \in (a, b)) = \int_a^b f_X(t) dt$$

- $f_X > 0 \forall x \in S$
- $\int_S f_X(x) dx = 1$ (normalization)
- $\forall A, P(X \in A) = \int_A f_X(x) dx$
- $F_X(a) = P(X \leq a) = \int_{-\infty}^a f_X(\tau) d\tau$

6.2 CDF

$$1. 0 \leq F_X(x) \leq 1, \forall x \in \mathbb{R}$$

2.

$$\lim_{x \rightarrow -\infty} F_X(x) = 0, \lim_{x \rightarrow \infty} F_X(x) = 1$$

$$3. a < b \Rightarrow F_X(a) \leq F_X(b)$$

$$4. P(a < X \leq b) = F_X(b) - F_X(a)$$

6.3 Expectation

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

$$1. \text{ If } f_X(x) \text{ is even symmetry about } m, E[X] = m$$

$$2. \text{ If } X \text{ is non negative RV,}$$

$$E[X] = \int_0^{\infty} (1 - F_X(x)) dx = \int_0^{\infty} P(X > x) dx$$

$$3. \text{ for any function } g \text{ where } g(x) \geq 0 \forall x,$$

$$E[g(X)] = \int_0^{\infty} (1 - F_g(x)) dx = \int_0^{\infty} P(g(X) > x) dx$$

6.4 Continuous Distributions

6.4.1 Continuous Uniform $X \sim U(a, b)$

$$f_X(x) = \frac{1}{b-a}, a < x < b \quad E[X] = \int_a^b x f_X(x) dx = \frac{b+a}{2}$$

$$E[X^2] = \int_a^b x^2 f_X(x) dx = \frac{b^2+ab+a^2}{3} \quad \text{Var}[X] = \frac{(b-a)^2}{12}$$

6.4.2 Gaussian/Normal RV $X \sim N(\mu, \sigma^2)$

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), x \in \mathbb{R}$$

$$\text{Conversion to standard gaussian } Z = \frac{X-\mu}{\sigma} \sim N(0, 1^2)$$

Non-correlation $\Leftrightarrow$ Independence of 2 Gaussian RV

6.4.3 Exponential $T \sim \text{Exp}(\lambda)$

$$f_T(t) = \lambda e^{-\lambda t}, t > 0 \quad F_T(t) = 1 - e^{-\lambda t}, t > 0$$

$$P(T > t) = 1 - F_T(t) = e^{-\lambda t}, t > 0 \quad E[T] = \frac{1}{\lambda}, \text{Var}[T] = \frac{1}{\lambda^2}$$

$$P(T > t_1 + t_2 | T > t_1) = P(T > t_2)$$

7 Chapter 6 Multiple Continuous random variable

7.1 Sum as Convolution

$$f_{X+Y}(z) = f_X(z) * f_Y(z) = \int_{-\infty}^{\infty} f_X(t) f_Y(z-t) dt$$

7.2 Central Limit Theorem

$$X_i \sim [\text{some distribution, with } \mu, \sigma^2] \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$Z_n = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{(\sum_{i=1}^n X_i) - n\mu}{\sqrt{n}\sigma} \text{ converges to standard } Z \text{ as } n \rightarrow \infty$$

7.3 Bivariate RV

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u, v) dv du \text{ and } F_X(x) = F_{X,Y}(x, \infty), F_Y(y) = F_{X,Y}(\infty, y)$$

$$P((x, y) \in A) = \iint_{(x,y) \in A} f_{X,Y}(x, y) dy dx$$

$$f_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y) = \frac{\partial^2}{\partial y \partial x} F_{X,Y}(x, y)$$

7.3.1 Bivariate Gaussian

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}}$$

$$\exp\left(-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho \frac{(x-\mu_X)}{\sigma_X} \frac{(y-\mu_Y)}{\sigma_Y} + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 \right]\right)$$

7.4 Continuity Correction

for bernoulli RV Y, ensure the boundary equality holds before correction

$$P(7 < X < 11) = P(8 \leq X \leq 10) = P(7.5 \leq X \leq 10.5)$$

7.5 Complex Gaussian

$$X \sim N(\mu_X, \sigma_X^2), Y \sim N(\mu_Y, \sigma_Y^2), H = X + iY$$

$$E[H] = \mu_X + i\mu_Y := \mu(\text{complex})$$

$$\text{var } \Gamma = E[|H - E[H]|^2] = \sigma_X^2 + \sigma_Y^2 := \sigma^2$$

7.5.1 Circularly Symmetric Gaussian RV

$$X \sim N(0, \sigma^2/2), Y \sim N(0, \sigma^2/2), H = X + iY \sim CN(0, \sigma^2) =$$

$$|H| \exp(i\angle H) = R \times \exp(i\theta) \text{ where}$$

$$R = |H| = \sqrt{X^2 + Y^2}, \theta = \angle H = \arctan(\frac{Y}{X}) \in [-\pi, \pi) \text{ (so that } (X, Y) \leftrightarrow (r, \theta) \text{ is bijective)}$$

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2+y^2}{2\sigma^2}\right)$$

$$P(A) = \int_A f_{X,Y}(x, y) dx dy = \int_A f_H(r, \theta) dr d\theta \text{ (note that } r \text{ is inside } f_H(r, \theta))$$

$$f_H(r, \theta + \phi) = f_H(r, \theta) = f_{X,Y}(x, y) r = \frac{r}{2\pi\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right)$$

$$f_R(r) = \int_{-\pi}^{\pi} f_H(r, \theta) d\theta = \frac{r}{\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right), r > 0$$

$$f_{\theta}(\theta) = \frac{1}{2\pi}, -\pi \leq \theta < \pi$$

8 problem solving techniques

8.1 Mapping of Random Variables

8.2 Derived Distribution

$$1. F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = \int_{\{x|g(x) \leq y\}} f_X(x) dx$$

$$2. f_Y(y) = \frac{dF_Y(y)}{dy}$$

If g is strictly monotonic and

$$\exists h \text{ s.t. } \forall x \in S_X, y = g(x) \Leftrightarrow x = h(y) = g^{-1}(y)$$

$$f_Y(y) = f_X(h(y)) \left| \frac{dh(y)}{dy} \right| > 0$$

8.3 Graphical Methods

area as probability